Zhenyu Zhang, Ph.D.

Research Scientist, iSEE, University of Illinois Urbana-Champaign
zhangzhy@illinois.edu | zhangzhy-geo.github.io | Google Scholar | ORCID 0000-0002-9155-7378

RESEARCH INTERESTS

Terrestrial water cycle and its coupling with the atmosphere, ecosystems, and agriculture, studied with process-based hydrological, agroecosystem, and Earth system models integrated with observations and AI: hydrometeorology and land–atmosphere interactions; coupled atmosphere–hydrology modeling (WRF / WRF-Hydro); ecohydrology and agroecosystem modeling (ecosys); model–data fusion; machine learning and agentic AI for hydrological and Earth system modeling; hydroclimatic extremes.

PROFESSIONAL APPOINTMENTS

Research Scientist, iSEE, University of Illinois Urbana-Champaign, USA — **2024 – present**
Visiting Scholar, UK Centre for Ecology & Hydrology (UKCEH), United Kingdom — **Nov. 2023**
Postdoctoral Research Associate, KIT IMK-IFU & University of Augsburg, Germany — **2019 – 2023**
Doctoral Researcher, Karlsruhe Institute of Technology (KIT), IMK-IFU, Germany — **2014 – 2019**

EDUCATION

Ph.D., Climate and Hydrology Science, University of Augsburg, Germany (advisor: H. Kunstmann) — **2020**
M.S., Earth System Science, Lanzhou University, China — **2014**
B.S., Geographic Information Science and Remote Sensing, Lanzhou University, China — **2011**

PUBLICATIONS

Peer-Reviewed Publications (* corresponding author)

32. **Zhang, Z.***, Guan, K.*, Grant, R., et al. (2026). Observation-constrained agroecosystem model inversion reveals continental-scale variation of winter wheat traits. *Global Change Biology*, 32(6), e70936. doi:10.1111/gcb.70936
31. Wang, X., Wu, J., Zhang, B., **Zhang, Z.**, et al. (2026). Reduced upwind moisture transport contributes to drought in the agro-pastoral ecotone of Northern China. *Geophysical Research Letters*, 53, e2025GL119587. doi:10.1029/2025GL119587
30. Shang, S., Ma, N., Zhu, G., Zhang, K., Chen, H., **Zhang, Z.**, et al. (2025). Global estimation of terrestrial evapotranspiration based on the atmospheric water balance approach. *Climate Dynamics*, 63, 64. doi:10.1007/s00382-024-07536-0
29. **Zhang, Z.***, Laux, P., Baade, J., et al. (2024). Sensitivity of joint atmospheric-terrestrial water balance simulations to soil representation: Convection-permitting coupled WRF-Hydro simulations for southern Africa. *Agricultural and Forest Meteorology*, 355, 110127. doi:10.1016/j.agrformet.2024.110127
28. **Zhang, Z.***, Laux, P., Baade, J., et al. (2024). Regional land–atmosphere interactions in southern Africa: Potential impact and sensitivity of forest and plantation change. In: von Maltitz, G.P., et al. (Eds.), *Sustainability of Southern African Ecosystems under Global Change*, Ecological Studies, vol. 248. Springer, Cham. (Book chapter) doi:10.1007/978-3-031-10948-5_10
27. Wei, J., Arnault, J., Rummler, T., Fersch, B., **Zhang, Z.**, et al. (2024). Acceleration of the hydrological cycle under global warming for the Poyang Lake Basin in Southeast China: An age-weighted regional water tagging approach. *Journal of Hydrometeorology*, 25, 1627–1647. doi:10.1175/JHM-D-23-0227.1
26. Shang, S., Zhu, G., Zhang, K., Chen, H., Wang, Y., Chen, Y., **Zhang, Z.**, et al. (2024). Spatial-temporal variations in evapotranspiration across the continental United States: An atmospheric water balance perspective. *Journal of Hydrology*, 640, 131699. doi:10.1016/j.jhydrol.2024.131699
25. Baade, J., Gessner, U., Hahndiek, E., Harmse, C., Hill, S., Hirner, A., Maruping-Mzileni, N., Otte, I., Pathe, C., Renner, P., Schellenberg, K., Selvaraj, S., Smith, C., Strydom, T., Swanepoel, A., Thonfeld, F., Urban, M., **Zhang, Z.**, et al. (2024). Observational support for regional policy implementation: Land surface change under anthropogenic and climate pressure in SALDi study sites. In: von Maltitz, G.P., et al. (Eds.), *Sustainability of*

Southern African Ecosystems under Global Change, Ecological Studies, vol. 248, pp. 845–877. Springer, Cham. (Book chapter) doi:10.1007/978-3-031-10948-5_29

24. **Zhang, Z.***, Laux, P., Baade, J., et al. (2023). Impact of alternative soil data sources on the uncertainties in simulated land-atmosphere interactions. *Agricultural and Forest Meteorology*, 339, 109565. doi:10.1016/j.agrformet.2023.109565
23. Arnault, J., Mwanthi, A.M., Portele, T., Li, L., Rummler, T., Fersch, B., Hassan, M.A., Bahaga, T.K., **Zhang, Z.**, et al. (2023). Regional water cycle sensitivity to afforestation: Synthetic numerical experiments for tropical Africa. *Frontiers in Climate*, 5, 1233536. doi:10.3389/fclim.2023.1233536
22. Wang, X., Zhang, B., **Zhang, Z.**, et al. (2023). Identifying spatiotemporal propagation of droughts in the agro-pastoral ecotone of northern China with long-term WRF simulations. *Agricultural and Forest Meteorology*, 336, 109474. doi:10.1016/j.agrformet.2023.109474
21. Wang, X., **Zhang, Z.**, Zhang, B., et al. (2023). Quantifying the impact of land use and land cover change on moisture recycling with convection-permitting WRF-tagging modeling in the agro-pastoral ecotone of northern China. *Journal of Geophysical Research: Atmospheres*, 128(8), e2022JD038421. doi:10.1029/2022JD038421
20. **Zhang, Z.***, Arnault, J., Laux, P., et al. (2022). Convection-permitting fully coupled WRF-Hydro ensemble simulations in high mountain environment: Impact of boundary layer- and lateral flow parameterizations on land-atmosphere interactions. *Climate Dynamics*, 59, 1355–1376. doi:10.1007/s00382-021-06044-9
19. Shang, S., Arnault, J., Chen, H., Wei, J., Zhang, K., **Zhang, Z.**, et al. (2022). Recent increase of spring precipitation over the Three-River Headwaters region — water budget analysis based on global reanalysis (ERA5) and ET-tagging extended regional climate modeling. *Journal of Climate*, 35(22), 3599–3617. doi:10.1175/JCLI-D-21-0829.1
18. Quenum, G.M.L.D., Arnault, J., Klutse, N.A.B., **Zhang, Z.**, et al. (2022). Potential of the coupled WRF/WRF-Hydro modeling system for flood forecasting in the Ouémé River (West Africa). *Water*, 14(8), 1192. doi:10.3390/w14081192
17. **Zhang, Z.***, Arnault, J., Laux, P., et al. (2021). Diurnal cycle of surface energy fluxes in high mountain terrain: High-resolution fully coupled atmosphere-hydrology modelling and impact of lateral flow. *Hydrological Processes*, 35(12), e14454. doi:10.1002/hyp.14454
16. Urban, M., Schellenberg, K., Morgenthal, T., Dubois, C., Hirner, A., Gessner, U., Mogonong, B., **Zhang, Z.**, et al. (2021). Using Sentinel-1 and Sentinel-2 time series for Slangbos mapping in the Free State Province, South Africa. *Remote Sensing*, 13(17), 3342. doi:10.3390/rs13173342
15. Laux, P., Dieng, D., Portele, T.C., Wei, J., Shang, S., **Zhang, Z.**, et al. (2021). A high-resolution regional climate model physics ensemble for Northern Sub-Saharan Africa. *Frontiers in Earth Science*, 9, 700249. doi:10.3389/feart.2021.700249
14. Arnault, J., Jung, G., Haese, B., Fersch, B., Rummler, T., Wei, J., **Zhang, Z.**, et al. (2021). A joint soil-vegetation-atmospheric modeling procedure of water isotopologues: Implementation and application to different climate zones with WRF-Hydro-Iso. *Journal of Advances in Modeling Earth Systems*, 13(10), e2021MS002562. doi:10.1029/2021MS002562
13. Shang, S., Zhu, G., Wei, J., Li, Y., Zhang, K., Li, R., Arnault, J., **Zhang, Z.**, et al. (2021). Associated atmospheric mechanisms for the increased cold season precipitation over the Three-River Headwaters region from the late 1980s. *Journal of Climate*, 34(19), 8033–8046. doi:10.1175/JCLI-D-21-0077.1
12. Wei, J., Dong, N., Fersch, B., Arnault, J., Wagner, S., Laux, P., **Zhang, Z.**, et al. (2021). Role of reservoir regulation and groundwater feedback in a simulated ground-soil-vegetation continuum: A long-term regional scale analysis. *Hydrological Processes*, 35(8), e14341. doi:10.1002/hyp.14341
11. Arnault, J., Fersch, B., Rummler, T., **Zhang, Z.**, et al. (2021). Lateral terrestrial water flow contribution to summer precipitation at continental scale – A comparison between Europe and West Africa with WRF-Hydro-tag ensembles. *Hydrological Processes*, 35(5), e14183. doi:10.1002/hyp.14183
10. **Zhang, Z.***, Arnault, J., Wagner, S., et al. (2019). Impact of lateral terrestrial water flow on land-atmosphere interactions in the Heihe River basin in China: Fully coupled modeling and precipitation recycling analysis. *Journal of Geophysical Research: Atmospheres*, 124(15), 8401–8423. doi:10.1029/2018JD030174
9. Arnault, J., Wei, J., Rummler, T., Fersch, B., **Zhang, Z.**, et al. (2019). A joint soil-vegetation-atmospheric water tagging procedure with WRF-Hydro: Implementation and application to the case of precipitation partitioning in the upper Danube River basin. *Water Resources Research*, 55, 6217–6243. doi:10.1029/2019WR024780

8. Arnault, J., Rummler, T., Baur, F., Lerch, S., Wagner, S., Fersch, B., **Zhang, Z.**, et al. (2018). Precipitation sensitivity to the uncertainty of terrestrial water flow in WRF-Hydro – An ensemble analysis for Central Europe. *Journal of Hydrometeorology*, 19, 1007–1025. doi:10.1175/JHM-D-17-0042.1
7. **Zhang, Z.**, Wang, N., Ma, N., et al. (2014). Lake area changes and the main causes in the hinterland of Badain Jaran Desert during 1973–2010, China. *Sciences in Cold and Arid Regions*, 6(1), 22–29. doi:10.3724/SP.J.1226.2014.00022
6. Ma, N., Wang, N., Zhao, L., **Zhang, Z.**, et al. (2014). Observation of mega-dune evaporation after different rain events in the hinterland of Badain Jaran Desert, China. *Chinese Science Bulletin*, 59(2), 162–170. doi:10.1007/s11434-013-0050-3
5. Wu, Y., Wang, N., Zhao, L., **Zhang, Z.**, et al. (2014). Hydrochemical characteristics and recharge sources of Lake Nuoertu in Badain Jaran Desert. *Chinese Science Bulletin*, 59(9), 886–895. doi:10.1007/s11434-013-0102-8
4. **Zhang, Z.**, Wang, N., Wu, Y., et al. (2013). Remote sensing on spatial changes of lake area in Badain Jaran Desert hinterland during 1973–2010. *Journal of Lake Sciences*, 25(4), 514–520. (In Chinese with English abstract) doi:10.18307/2013.0408
3. Li, Z., Wang, N., Li, Y., **Zhang, Z.**, et al. (2013). Runoff simulations using water and energy balance equations in the lower reaches of the Heihe River, northwest China. *Environmental Earth Sciences*, 70, 1–12. doi:10.1007/s12665-012-2099-8
2. Wang, N., Ma, N., Chen, H., Chen, X., Dong, C., **Zhang, Z.** (2013). Preliminary study for the precipitation characteristics in the hinterland of Badain Jaran Desert. *Advances in Water Science*, 24(2), 153–160. (In Chinese with English abstract) http://skxjz.nhri.cn/article/id/50
1. **Zhang, Z.**, Wang, N., Ma, N., et al. (2012). Lakes area change in Badain Jaran Desert hinterland and its influence factors during the recent 40 years. *Journal of Desert Research*, 32(6), 1743–1750. (In Chinese with English abstract) http://www.desert.ac.cn/CN/Y2012/V32/I6/1743

RESEARCH PROJECTS

Scientist, U.S. DOE Center for Advanced Bioenergy and Bioproducts Innovation (CABBI)	**2024 – present**
PI, Open Fund, Key Laboratory of Geospatial Technology, Henan University, China	**2022 – 2023**
Co-I, European Space Agency (ESA), EO AFRICA R&D Facility (awarded, not executed)	**2022**
Co-I, BMBF–MoST joint project on compound extreme flooding in coastal cities	**2021 – 2024**
Scientist, BMBF SPACES II, South Africa Land Degradation Monitor (SALDi)	**2019 – 2023**

PRESENTATIONS

2026	Phenology-aligned climate attribution reveals dominant overwinter controls on U.S. winter wheat. *AGU Fall Meeting 2026*, San Francisco, CA. (Oral)
2026	Observation-constrained agroecosystem model inversion links winter wheat trait diversity to climate vulnerability. *AGU Fall Meeting 2026*, San Francisco, CA. (Oral)
2024	Assessing winter wheat productivity and carbon budgets across the United States using agroecosystem modeling. *AGU Fall Meeting 2024*, Washington, D.C. (Poster)
2023	Regional coupled atmospheric-land surface-hydrological modeling for South Africa: Development, uncertainties and challenges. *SA–DE Workshop: Ocean–Climate–Land Interactions and Feedbacks under Climate Change*, Thünen Institute, Braunschweig, Germany. (Oral)
2022	Fully coupled atmospheric-hydrological modeling in the investigation of water balance and land-atmosphere interactions. *Institute of Tibetan Plateau Research, Chinese Academy of Sciences*, Beijing, China. (**Invited**)
2022	Improved understanding of regional water cycle using a coupled atmospheric-hydrological modeling framework. *2nd Bilateral Korea–Germany Collaboration Initiation, Korea University*, Seoul, South Korea. (**Invited**)
2022	Presentation of a high-resolution present-day joint atmosphere-land surface-hydrology simulation dataset for South Africa. *EGU General Assembly 2022*, Vienna, Austria. (Oral)

2022 Presentation of a high-resolution present-day joint atmosphere-land surface-hydrology simulation dataset for South Africa. *Savanna Science Network Meeting 2022*, Skukuza, South Africa. (Poster)

2021 Evaluating the role of soil physical properties on simulated land-atmosphere interactions over South Africa using coupled atmosphere-hydrological modeling. *EGU General Assembly 2021*, online. (Oral)

2020 Assessing the interactions of atmosphere and land surface over South Africa with convection-permitting coupled atmospheric-hydrological modeling. *EGU General Assembly 2020*, online. (Online display)

2019 Sensitivity of convection-permitting atmospheric-hydrological modeling to PBL parameterization in simulating land-atmosphere interaction over the Heihe River basin, northwest China. *EGU General Assembly 2019*, Vienna, Austria. (Poster)

2018 Joint atmospheric-terrestrial water cycle and land-atmosphere interactions in complex terrain: A case study for the Heihe River basin. *Sino-German Fall School "Modelling Platform Prototype for Environmental System Dynamics"*, Nanchang, China. (**Invited**)

2018 Effect of lateral terrestrial water flow on land-atmosphere feedbacks: A comparison between WRF and WRF-Hydro simulations for the Heihe River basin in China. *EGU General Assembly 2018*, Vienna, Austria. (Oral)

2017 Enhancement of hydrological parameterization and its impact on atmospheric modeling: A WRF-Hydro case study in the upper Heihe River basin, China. *EGU General Assembly 2017*, Vienna, Austria. (Poster)

SERVICE AND MENTORING

- Review Editor and Guest Editor, *Frontiers in Water* (2023 – present)
- Reviewer of 100+ manuscripts for 25+ journals, including *Journal of Hydrology*, *Water Resources Research*, *HESS*, *Journal of Hydrometeorology*, *Agricultural and Forest Meteorology*, *JGR: Atmospheres*, and *Journal of Climate*
- Co-mentored Ph.D. and visiting students from China and Africa, KIT IMK-IFU & University of Augsburg (2019 – 2023)
- Member, American Geophysical Union (2024 – present); European Geosciences Union (2017 – 2022)

Last updated: October 2026